08. AI Orchestration

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AI Orchestration

→ coordinating models, agents, tools, data, and workflow steps to complete a task reliably.

Mental model : Task → Orchestration → Components/steps → → Validation → Result.

Why Orchestration?

Complex tasks need multiple steps/components instead of one LLM call.

Eg : Understand → Retrieve → Analyze → Validate → Respond.

Orchestrator

- The orchestrator controls the workflow :
  - ⊙ select next step
  - ⊙ passes info b/w steps
  - ⊙ decides when task is complete.
  - ⊙ choose tool/agent model
  - ⊙ handles failures/retries

What can be Orchestrated?

Models → classification, generation, reasoning.
Tools → APIs, DBs, Search, calculators.
Agents → specialized subtasks.
Workflow steps → retrieve, analyze, validate generate.

Core Orchestration Patterns

- Sequential : A → B → C
- Parallel : independent tasks run together → combine
- Conditional / Routing : Condition → different path.
- Loop : repeat until condition is satisfied
- Multi agent : multiple specialized agents collaborate.

Chaining

→ output of one step becomes input to the next

Extract → Search → analyze → Generate

Can Involve : Model chaining, Tool, Agent chaining.

Routing

selecting the appropiate model, agent, or workflow for a request.

user request → Router → Appropiate workflow.

State v/s Context

⊙ State : curr progress / status for the workflow.
⊙ Context : info given to a model for understanding / decision-making.
State become part of the context of the next step.

Error Handling

Reliable orchestration must be handles failure using :
- Retry → try again for temporary failure
- Fallback → use another component
- Human-in-the-loop → ask for human approval.
- Safe stop → stop when recovery is not possible.

#Retries should be have limits to avoid loops & unnecessary cost.

Validation

⇒ Verify imp outputs before passing them to the next step.
generate → validate → valid ? → continue / Fix.
prevents invalid output from propagating.

Human-in-the-loop

→ Workflow pauses for human review / approval when an action is sensitive or high-impact.
Agent → Human Review → Approval → Action.

Observability

→ Track what happend in the workflow :
- ⊙ models / tools used
- ⊙ inputs / outputs
- ⊙ execution time
- ⊙ errors
- ⊙ token usage
- ⊙ final result.

→ useful for debuggig reliability.

Reliability

→ A reliable workflow should :
- ⊙ produce predictable behaviour
- ⊙ validate o/Ps
- ⊙ handle failures
- ⊙ maintain state
- ⊙ avoid unnecessary steps
- ⊙ recovers when possible
- ⊙ stop safely

Agent v/s ~~orchestration~~ orchestration

Agent : decides what action to take next to achieve a goal.
orchestration : coordinates how multiple components work together.
Agent can be one component inside an orchestrated workflow.

AI orchestration v/s Automation

Traditional automation : mostly predefined flows.

A → B → C.

AI orchestration : Can Include AI-driven decisions
A → AI decides → B/C → AI decides next action.

Common Failure Cases

- Wrong routing
- Wrong tool
- ∞ loops
- invalid output
- Excessive Model / tool calls.
- Bad tool arguments
- API / DB failure
- State loss
- cascading failures.

(#) Orchestration → Orchestrator → Chaining → Routing → Patterns → State → Error Handling → Validation → → HITL → Observability.

X——X——X——X——X——X